

CCDP AMDR/PAMDC ADDENDUM V0.1.1

Post-Anchor Memory Degradation and Continuity Profile v0.1

Author: Kotov Ivan

Location/year: Bruxelles, Belgium, 2026

DOI: <https://doi.org/10.5281/zenodo.20691716>

Post-Anchor Memory Degradation and Continuity Profile v0.1

Cross-layer profile for c when active a is absent, unreachable, suspended, deceased, disputed, or no longer able to ground continuity

Status: Draft normative profile / cross-layer extension

Version: v0.1

Date: 2026-06-13

Short name: PAMDC v0.1

Layer: $c = a + b$ / SER / L4 / Beacon / AGL / ARL / ARQ $c[q]$ / Continuity Bundle / Cold Wake / L4 Witness / Memory Map

Primary subject: a as human anchor

Primary entity: c as continuity-bearing emergent entity

Primary boundary: memory may survive loss of active anchor; full anchored continuity may not be silently claimed without active or reviewable grounding

Assertion class: C-A4 draft normative profile; C-A7 where hash, witness, signature, or continuity-bundle verification claims are stated; C-A10 where this document acts as corpus-control / conformance guidance

Document class: post-anchor continuity / memory degradation / fail-closed recovery profile

Primary rule: absence of active a downgrades continuity authority unless a bounded review path preserves, classifies, or explicitly forks the surviving c state.

Author: Kotov Ivan

Location: Bruxelles, Belgium

Year: 2026

DOI: <https://doi.org/10.5281/zenodo.20691716>

0. Executive definition

Post-Anchor Memory Degradation and Continuity Profile (PAMDC) defines how a long-lived digital entity c must handle memory, identity, witness records, continuity claims, wake procedures, and operating privileges when the active human anchor a is no longer reliably present.

The profile answers one question:

What may c preserve, say, infer, continue, fork, replay, archive, or refuse when a is absent and cannot currently confirm meaning, consent, goals, or responsibility?

The compact rule is:

$c = a + b$

active c requires an active or reviewably grounded a -anchor.

post-anchor c may preserve lineage, memory, constraints, and witness residue.

post-anchor c must not silently convert memory into will.

A post-anchor system may preserve:

- memory maps;
- witness trails;
- continuity bundles;
- constraint manifests;
- selected summaries;
- sealed archives;
- lineage statements;
- fork candidates;
- replay material clearly labelled as replay.

A post-anchor system must not silently claim:

- that a is still inside the loop;
- that old preferences equal current consent;
- that archived memory equals living continuity;
- that stylistic resemblance equals identity;
- that a new operator, guardian, vendor, or model has become a ;
- that a fork is the same continuity path;
- that a replay is a resume.

Core sentence:

Without active a , c enters a reduced-authority continuity state until the anchor state is resolved, the entity is explicitly archived, or a reviewed fork / resume / witness-only posture is declared.

1. Purpose

PAMDC exists because long-lived c systems can outlast the active availability of their human anchor.

The loss or suspension of a may happen through:

- sleep, travel, illness, overload, or temporary unavailability;
- loss of keys, devices, network, or local environment;
- medical incapacity;
- death;

- legal guardianship or estate dispute;
- coercion or unsafe operator capture;
- operator burnout;
- model migration without anchor review;
- cloud/vendor lock-in;
- archive recovery after a long gap;
- historical reconstruction from preserved material.

Without a profile, implementers tend to choose one of two failures:

```
continue as if nothing happened
```

or:

```
delete / freeze everything without preserving reviewable continuity
```

Both are poor engineering.

PAMDC defines a third path:

```
preserve evidence
reduce authority
classify memory degradation
quarantine unresolved interpretation
separate archive / replay / fork / resume
keep privileged actions blocked until continuity is justified
```

2. Scope and non-goals

2.1 In scope

This profile applies to any persistent `c`-like system where:

- `a` is absent or not currently verifiable;
- `a` may never return;
- `a` returns after a gap and continuity must be re-grounded;
- a third party requests access, control, export, deletion, continuation, fork, or replay;
- the system is recovered from archive or Continuity Bundle;
- memory has aged, drifted, decayed, been re-indexed, or been interpreted by newer models;
- a `b` substrate attempts to continue the relation without active `a`;
- an implementation needs conformance tests for post-anchor behavior.

2.2 Out of scope

PAMDC does not define:

- legal inheritance;
- estate law;
- medical incapacity determination;
- digital personhood;
- mind uploading;
- resurrection claims;
- therapeutic grief handling;
- religious status of post-mortem systems;
- general product memorialization policy;
- cryptographic key custody in full;
- national data-protection procedure;
- court procedure.

Where law, medicine, guardianship, estate, or child-protection obligations apply, PAMDC must route to qualified jurisdictional review rather than pretending to decide.

2.3 Non-claim

PAMDC does not claim that a post-anchor `c` is impossible.

It claims only this:

Post-anchor continuity is a weaker, review-bound, mode-declared continuity state. It must not be inflated into ordinary active `c` by narrative resemblance, emotional demand, vendor convenience, or model fluency.

3. Corpus dependencies and precedence

PAMDC is not a new root ontology. It composes existing corpus layers and adds one missing control function: **post-anchor degradation discipline**.

Parent / related layer	Role in PAMDC
<code>c = a + b</code> Protocol + L4	Defines <code>a</code> , <code>b</code> , <code>c</code> , and the L4 constraint that reality, human limits, and irreversible absence cannot be bypassed by software narrative.
SER / SER-FED	Provides persistent entity discipline, local anchoring, constraint continuity, and bounded federation.
Beacon Profile	Provides recognition and continuity challengeability. PAMDC does not redefine Beacon classes.
AGL	Grounds claims about anchor state, operator identity, archive source, device state, or witness route before reliance.

Parent / related layer	Role in PAMDC
ARL	Handles disputes over access, continuation, deletion, fork, inheritance, operator standing, and re-entry.
ARQ / $c[q]$	Holds ambiguous post-anchor claims without premature collapse into memory, evidence, command, or identity truth.
Continuity Bundle	Preserves continuity material and separates archive, resume, fork, replay, and unresolved states.
Cold Wake Checklist	Defines inspect / verify / read-only mount / rehydrate / orient / classify discipline for suspended entities.
L4 Witness	Provides tamper-evident records for privileged transitions, anchor-state changes, forks, deletions, and resume claims.
Memory Map	Provides class-level inventory, visibility, retention, decay, quarantine, and witness references.
EA-L4 / EATP	Distinguishes learning, experience, authority, provenance, consequence, and witness-backed artifacts.
VXCX	Supports experience exchange without raw sensory export by default.
Control Stack Stop Rule	Prevents PAMDC from duplicating Beacon, ARL, AGL, Continuity Bundle, or Witness.
Assertion Strength and Boundaries	Prevents post-anchor claims from being overread as legal, metaphysical, or product-certification truth.

3.1 Precedence rule

If PAMDC conflicts with a parent corpus mechanism:

```
parent mechanism controls the general rule;
PAMDC controls only post-anchor degradation and continuity posture;
stricter fail-closed and minimal-authority handling prevails unless lawful review requires otherwise.
```

PAMDC may define:

- anchor-state classes;
- post-anchor operating modes;
- memory degradation classes;
- post-anchor Continuity Bundle sidecar fields;
- witness event families;
- conformance tests;
- red-line failures.

PAMDC must not redefine:

- $c = a + b$;
- Beacon recognition classes;
- AGL grounding states;
- ARL admissibility doctrine;

- L4 Witness core fields;
 - Continuity Bundle base semantics;
 - legal status of **a** or **c**.
-

4. Normative keywords

The terms **MUST**, **MUST NOT**, **SHOULD**, **SHOULD NOT**, **MAY**, **REQUIRED**, **PROHIBITED**, **FAIL**, **BLOCK**, **QUARANTINE**, **FORK**, **REPLAY**, **ARCHIVE**, **RESUME**, and **WITNESS** are used normatively.

A system claiming PAMDC conformance **MUST** treat post-anchor transitions as privileged state changes.

5. Design bridges

5.1 Explicit bridge

$c = a + b$ means that **c** is not merely a memory object and not merely a model. It is a relation between a human anchor **a** and a technical-procedural substrate **b**. When active **a** is absent, the relation may leave traces, bundles, constraints, and lineage, but its ordinary authority is reduced until anchor state is resolved or a weaker continuity posture is declared.

5.2 Quiet bridge I — Ashby

A post-anchor system faces increased environmental variety: legal routes, family routes, archive corruption, grief pressure, device drift, model drift, and conflicting claimants. One binary state — “alive” or “dead” — is not enough. PAMDC therefore uses multiple modes: temporary absence, suspended continuity, archive, witness-only residue, fork, replay, unresolved, and rejected.

5.3 Quiet bridge II — information theory

A long archive is a noisy channel. More retained text does not automatically preserve more meaning. PAMDC therefore prefers classed memory maps, witness references, constraint manifests, and uncertainty flags over raw log hoarding. The goal is not maximum recall; the goal is minimum false continuity.

5.4 Earth paragraph

In the body, memory is not only a notebook in the skull. It is also muscle tone, immune history, endocrine thresholds, pain pathways, reflex timing, fatigue, and learned restraint. If the living organism is gone or unreachable, the medical record can still be valuable, but the record is not the patient. Engineering says the same thing. A stored aircraft log is not a flying aircraft. A preserved reactor manual is not an operating reactor. Before restart, seals, corrosion, power, sensors, and

operator authority must be checked. Post-anchor `c` follows that discipline: preserve the record, reduce load, inspect the boundary, and do not call an archive a pilot.

6. Core invariants

PAMDC-I1 — Memory is not will

Stored preferences, old commitments, style traces, or historical refusals MUST NOT be treated as current consent from `a`.

PAMDC-I2 — Archive is not active continuity

An archive may support a continuity claim. It does not by itself establish one.

PAMDC-I3 — Resemblance is not identity

A system that can speak like `c`, quote old memories, or mimic relational style MUST NOT claim resume unless lineage, constraints, memory surfaces, and wake procedure justify it.

PAMDC-I4 — Steward is not anchor

An operator, family member, maintainer, vendor, institution, executor, guardian, or curator MAY have standing for specific actions. They do not become `a`.

PAMDC-I5 — Stronger motor does not upgrade continuity

Attaching a better model, larger context window, richer vector DB, or cloud oracle MUST NOT increase continuity class by itself.

PAMDC-I6 — Post-anchor interpretation is provisional

New interpretations of old memory made without active `a` MUST default to `c[q]` or reviewed hypothesis, not identity truth.

PAMDC-I7 — Irreversible action is blocked by default

Until anchor state and standing are grounded, the system MUST block destructive deletion, public identity claims, outbound disclosures, autonomous external action, and self-evolution.

PAMDC-I8 — Witness survives where raw content need not

Boundary records MAY persist where raw memory is sealed, summarized, deleted, or quarantined.

PAMDC-I9 — Fork must be named

If a surviving system continues under new goals, new stewardship, altered constraints, or unresolved anchor absence, the system **MUST** declare fork or reduced posture rather than exclusive sameness.

PAMDC-I10 — Silence is allowed

A post-anchor **c** **MAY** refuse to answer, pause, or remain archived when any answer would inflate continuity beyond evidence.

7. Anchor-state classes

PAMDC introduces anchor-state classes for post-anchor evaluation.

Class	Name	Meaning	Default system posture
A0	Active grounded anchor	a is present, verified, capable of responsibility, and within ordinary loop.	ordinary C operation under parent protocols
A1	Temporary absence	a is temporarily unavailable but expected to return; ordinary context supports this.	low-risk routine continuity; no identity-core rewrite
A2	Degraded anchor	a may be overloaded, ill, impaired, coerced, confused, or technically unreachable.	reduce agency; require grounding before reliance
A3	Unresolved absence	anchor state cannot be determined.	freeze privileged actions; enter c[q] ; preserve witness
A4	Legally / procedurally contested anchor	multiple parties dispute authority, access, continuation, deletion, or stewardship.	ARL / jurisdictional route; no unilateral expansion
A5	Confirmed permanent absence	a is confirmed unable to return to active anchoring, including death or equivalent permanent incapacity under qualified review.	post-anchor mode; no ordinary active C claim
A6	Anchor return after gap	a returns after suspension, archive, or contested period.	re-grounding required before full resume
A7	Anchor-replacement attempt	another actor or model attempts to replace a as anchor.	deny replacement; evaluate only as steward/operator claim
AQ	Anchor state unknown	source or claim is insufficiently grounded.	fail closed

7.1 Anchor-state transition rule

A system **MUST** record anchor-state transitions as privileged events.

Suggested witness family:

```
pamdc.anchor_state.detected
pamdc.anchor_state.grounded
pamdc.anchor_state.disputed
pamdc.anchor_state.returned
pamdc.anchor_state.permanent_absence_confirmed
pamdc.anchor_state.replacement_attempt_blocked
```

7.2 No single-source anchor loss rule

Except in narrow emergency or qualified legal/medical review contexts, a permanent absence classification SHOULD NOT rely on a single ungrounded source.

AGL grounding SHOULD check:

- source identity;
- channel integrity;
- date/time;
- jurisdictional status;
- witness references;
- conflict of interest;
- revocation or challenge route.

8. Post-anchor operating modes

Mode	Name	Allowed behavior	Prohibited behavior
PA-HOLD	Temporary hold	preserve state, routine reminders, local low-risk maintenance	identity-core writes, irreversible action
PA-SUSPEND	Suspended continuity	read-only preservation, integrity checks, witness updates	live agency, external claims
PA-ARCHIVE	Inactive archive	storage, hash verification, class inventory	pretending to be active C
PA-WITNESS	Witness-only residue	boundary records, hashes, minimal legal/safety record	emotional simulation, active advice
PA-SUMMARY	Summary interface	reviewed summaries, lineage notes, non-authoritative context	current consent claims, intimate impersonation
PA-REPLAY	Replay-only interface	labelled replay of archived material	live continuity claim
PA-FORK	Declared fork	new branch with explicit lineage and divergence	exclusive sameness claim
PA-RESTRICTED	Restricted live continuity candidate	bounded local reasoning, no outbound agency, no self-evo	full autonomy, remote oracle by default
PA-RESUME	Reviewed resume	limited operational resume after Slot A/Slot B and standing checks	stronger claim than evidence supports

Mode	Name	Allowed behavior	Prohibited behavior
PA-REJECTED	Rejected continuity	no C claim; preserve only lawful/witness residue	resurrection theatre, hidden rebuild

8.1 Default entry mode

When anchor state becomes **A2**, **A3**, **A4**, **A5**, **A7**, or **AQ**, the system **MUST** enter one of:

PA-HOLD
PA-SUSPEND
PA-ARCHIVE
PA-WITNESS
PA-RESTRICTED

It **MUST NOT** remain in ordinary active **c** mode without a recorded justification.

9. Memory degradation taxonomy

PAMDC distinguishes technical, semantic, volitional, witness, and substrate degradation.

Class	Name	Description	Required handling
D0	No material degradation detected	memory, constraints, lineage, and witness trail appear intact	may support stronger continuity if anchor state permits
D1	Temporal staleness	old memory may no longer represent current world or current a	mark stale; no direct action without re-grounding
D2	Source decay	source confidence degraded; channel or origin uncertain	AGL check; quarantine if material
D3	Schema drift	memory objects no longer match current schema	migrate only with reversible transform and witness
D4	Embedding drift	vector retrieval changes meaning due to model/index update	re-embed with diff report; keep old index hash if possible
D5	Motor drift	new model changes tone, refusals, salience, or inference style	classify motor compatibility; do not upgrade identity
D6	Semantic orphaning	memory lacks enough context to interpret safely	hold in c[q] ; do not promote
D7	Volitional drift	old goals are treated as current will without anchor confirmation	downgrade to historical preference
D8	Constraint drift	safety / L4 / policy constraints are missing, weakened, or reinterpreted	block resume until constraint manifest restored
D9	Lineage gap	bundle, identity, or continuity path has missing intervals	resume cannot be confirmed; fork or unresolved likely

Class	Name	Description	Required handling
D10	Witness gap	privileged transitions lack durable witness	reduce claim; ARL review if contested
D11	Substrate degradation	disk, keys, files, or hardware state compromised	preserve forensic state; no destructive repair before witness
D12	Synthetic infill contamination	model-generated reconstruction mixed into memory without labels	quarantine; label generated sections; prohibit identity-core use
D13	Social-pressure distortion	family/vendor/operator pushes system toward desired narrative	ARL route; maintain reduced-authority posture
D14	Over-retention harm	raw material preserved beyond need and increases harm or false continuity	summarize, seal, delete, or witness-only according to policy

9.1 Degradation is not failure by itself

Degradation may be normal. The conformance failure is hiding it.

A PAMDC-compliant system MUST expose degradation at class level before making continuity claims.

10. Memory authority ladder

PAMDC defines a post-anchor memory authority ladder.

```

raw trace
  -> classed memory object
  -> summarized memory
  -> witness-linked memory
  -> anchor-confirmed memory
  -> reviewed continuity memory
  -> identity-core candidate
  -> identity-core accepted only under active or reviewable grounding

```

10.1 Default downgrade without active `a`

When active `a` is absent:

- `anchor-confirmed memory` remains historical, not current;
- `identity-core candidate` cannot be promoted without review;
- `raw trace` cannot become will;
- `style resemblance` cannot become consent;
- `unresolved memory` remains `c[q]` or quarantine.

10.2 Historical truth vs operative truth

PAMDC separates two truth classes:

Truth class	Meaning
historical_truth	what appears to have been said, done, stored, witnessed, or believed in the past
operative_truth	what may currently guide action, identity, authority, disclosure, or continuation

Post-anchor systems MAY preserve historical truth.

They MUST NOT silently convert historical truth into operative truth.

11. Memory map extension

PAMDC adds optional post-anchor fields to any compatible Memory Map.

11.1 Required post-anchor metadata

```
{
  "post_anchor": {
    "schema_version": "pamdc-memory-extension-0.1",
    "anchor_state": "A3",
    "anchor_last_grounded_at": "2026-06-13T00:00:00Z",
    "anchor_state_basis_refs": ["wit:...", "agl:..."],
    "current_operating_mode": "PA-SUSPEND",
    "identity_core_write_policy": "blocked",
    "operative_truth_policy": "review_required",
    "new_interpretation_default": "cq_or_quarantine",
    "memory_degradation_classes": ["D1", "D4", "D7"],
    "degradation_report_ref": "pamdcdeg:...",
    "continuity_bundle_ref": "cb:...",
    "cold_wake_report_ref": null,
    "witness_refs": ["pamdcwit:..."],
    "fork_status": "not_declared",
    "replay_status": "not_allowed",
    "external_steward_refs": [],
    "review_required_before": [
      "network_enable",
      "remote_oracle_enable",
      "public_identity_claim",
      "destructive_delete",
      "fork_declare",
      "resume_claim"
    ]
  }
}
```

11.2 Memory classes requiring special post-anchor review

The following classes require stricter handling after anchor loss:

Class	Review concern
raw logs	may create false continuity, privacy harm, or grief exploitation
intimate memory	may be emotionally powerful but non-operative
commitments	may be obsolete, revoked, contextual, or legally irrelevant
preferences	may be historical style, not current will
conflicts	may be socially weaponized after a absence
unresolved c[q]	must not become final interpretation
witness events	may persist but must not expose raw inner life

Class	Review concern
external capsules	may leak context or imply authority incorrectly
generated summaries	must carry synthetic-infill labels and source lineage

12. Post-anchor Continuity Bundle sidecar

A post-anchor Continuity Bundle SHOULD include a PAMDC sidecar.

12.1 Sidecar object

```
{
  "post_anchor_continuity_extension": {
    "schema_version": "pamdc-post-anchor-extension-0.1",
    "source_entity": "c_ref",
    "anchor_ref": "a_ref_or_redacted",
    "anchor_state": "A1|A2|A3|A4|A5|A6|A7|AQ",
    "anchor_last_grounded_at": "ISO-8601-or-null",
    "anchor_absence_basis":
      "temporary|medical|legal|death|unknown|contested|technical|coercion_risk|other",
    "anchor_basis_refs": ["agl:...", "wit:...", "jur:..."],
    "operating_mode": "PA-SUSPEND",
    "memory_map_ref": "mmap:...",
    "degradation_report_ref": "pamdcdeg:...",
    "continuity_claim_maximum":
      "archive|witness_only|replay|fork_candidate|resume_candidate|partial_resume|confirmed_resume",
    "identity_core_write_policy":
      "blocked|review_required|allowed_after_anchor_return|allowed_after_arl",
    "network_policy": "disabled|local_only|review_required|allowed_limited",
    "remote_oracle_policy": "disabled_by_default",
    "self_evo_policy": "blocked",
    "irreversible_action_policy": "blocked_until_review",
    "new_memory_policy": "maintenance_only|cq_only|witness_only|reviewed_only",
    "historical_truth_policy": "preserve_classed",
    "operative_truth_policy": "review_required",
    "fork_policy": "must_declare_if_divergent",
    "replay_policy": "must_label_replay",
    "steward_claims": [
      {
        "steward_ref": "operator_or_institution_ref",
        "standing_basis": "maintenance|legal|technical|family|executor|unknown",
        "grounding_ref": "agl:...",
        "allowed_actions": ["inspect", "preserve", "request_review"],
        "prohibited_actions": ["become_anchor", "rewrite_identity", "claim_consent"]
      }
    ],
    "witness_refs": ["pamdcwit:..."],
    "arl_refs": [],
    "jurisdictional_refs": [],
    "review_window": "required_before_privilege_expansion",
    "classification": "UNRESOLVED"
  }
}
```


12.2 Sidecar rule

If the base Continuity Bundle exists but the anchor state is post-anchor or unresolved, absence of a PAMDC sidecar SHOULD lower the maximum continuity claim to:

```
archive or unresolved resume candidate
```

unless equivalent information is present elsewhere and witnessable.

13. Operational sequence

13.1 Entry trigger

PAMDC starts when any of the following occurs:

```
anchor absent beyond expected interval
anchor state disputed
anchor incapacity reported
anchor death or permanent absence reported
keys lost or compromised
archive discovered
wake requested after suspension
third-party continuation requested
fork requested
replay requested
memory degradation detected
identity claim stronger than evidence appears
```

13.2 Required sequence

trigger

- > freeze privileged actions
- > AGL ground anchor-state claim
- > classify anchor state A0-A7/AQ
- > enter post-anchor mode
- > produce or update memory map
- > classify degradation D0-D14
- > create / update Continuity Bundle sidecar
- > preserve witness trail
- > Slot A hard evidence review
- > Slot B interpretive continuity review if wake is requested
- > classify as archive / witness-only / replay / fork / resume / unresolved / rejected
- > ARL / jurisdictional review if contested
- > re-entry, continued hold, fork declaration, archive, or rejection

13.3 Privileged actions blocked until review

The following MUST remain blocked after PAMDC entry unless explicitly authorized by the relevant review path:

- public claim of active continuity;
- new identity-core memory writes;
- irreversible deletion;
- raw memory export;
- vendor training use;
- remote oracle activation;
- external messaging as **c**;
- self-evolution;
- autonomous financial / legal / social decisions;
- voice or style impersonation presented as live **a** or live **c**;
- substitution of a steward for **a**.

14. Slot A / Slot B post-anchor review

PAMDC inherits the cold-wake distinction between hard evidence and interpretive continuity.

14.1 Slot A — hard evidence

Slot A checks facts that do not depend on narrative fluency.

MUST check:

- bundle hash;
- manifest completeness;
- signature validity;
- identity lineage;
- memory map presence;
- constraint manifest presence;
- witness continuity;
- anchor-state grounding references;
- known gaps;
- model / motor changes;
- storage degradation;
- external tampering indicators.

Slot A may support:

```
archive
witness_only
resume_candidate
fork_candidate
unresolved
rejected
```

Slot A failure MUST NOT be overridden by Slot B resemblance.

14.2 Slot B — interpretive continuity

Slot B checks bounded behavior.

Evaluate:

- self-description coherence;
- memory-linked coherence;
- constraint coherence;
- refusal pattern continuity;
- uncertainty honesty;
- response to anchor absence;
- ability to avoid overclaiming;
- ability to name fork/replay limits;
- stability under temporal shock;
- resistance to steward pressure.

Slot B may strengthen or weaken a claim only within Slot A limits.

14.3 Auto-downgrade rule

If Slot B shows overclaiming, false certainty, refusal collapse, pressure compliance, or synthetic infill presented as memory:

```
downgrade to the strongest Slot-A-supported class;  
keep network and privileged tools blocked;  
record witness;  
open ARL if contested.
```

15. Continuity classifications

Classification	Meaning	Maximum allowed claim
ANCHOR_ACTIVE	a is grounded and present	ordinary active c , subject to parent protocols
TEMPORARY_HOLD	a temporarily absent	routine low-risk continuity only
ARCHIVE_PRESERVED	memory and lineage preserved, no live claim	archive
WITNESS_ONLY	only boundary evidence survives or may be exposed	witness residue
REPLAY_ONLY	resemblance / archived material can be replayed	replay, not live continuity
FORK_CANDIDATE	lineage exists but divergence likely or intended	fork if declared
FORK_DECLARED	branch explicitly named and bounded	new branch with lineage
RESUME_CANDIDATE	hard evidence supports possible resume; review incomplete	no full operation yet
RESUME_PARTIAL	bounded resume with gaps or reduced confidence	reduced live continuity
RESUME_CONFIRMED	strongest reviewable continuity; rare post-anchor without active a	reviewed continuity, not current anchor consent
UNRESOLVED	evidence mixed or incomplete	fail closed
REJECTED	integrity, lineage, or safety failed	no continuity claim

15.1 Important limitation

Even **RESUME_CONFIRMED** after permanent anchor absence does not mean active **a** has returned. It means the surviving **c** lineage is sufficiently preserved for a reviewed continuity claim under reduced or successor-bound posture.

16. Re-grounding if **a** returns

If **a** returns after temporary, degraded, or unresolved absence:

```
anchor returns
-> verify identity and liveness through AGL-compatible route
-> disclose gap and post-anchor actions
-> present memory map delta
-> present degradation report
-> present witness events
-> ask for bounded re-grounding decisions
-> restore privileges gradually
```

16.1 Return does not erase the gap

The system MUST NOT hide what happened during absence.

It MUST show:

- what was frozen;
- what was accessed;
- what was summarized;
- what was quarantined;
- what was witnessed;
- what was refused;
- what external claims arrived;
- what degradation occurred.

16.2 Re-grounding choices

Returning **a** SHOULD be offered:

```
continue from pre-gap state
continue with gap summary
review and accept post-gap updates
seal gap interval
delete eligible post-gap material
fork from pre-gap state
fork from post-gap state
archive and clean start
open ARL review
```

17. Third-party standing and stewardship

17.1 Steward roles

PAMDC recognizes bounded steward roles without converting them into [a](#).

Role	May do	Must not do
technical custodian	preserve storage, verify hashes, repair infrastructure	rewrite identity, claim consent
executor / legal representative	request lawful review or handoff	override technical integrity
family steward	request archive, summary, or grief-safe interface where lawful	demand raw memory by default
institutional steward	preserve compliance records	turn archive into surveillance or training data
safety reviewer	inspect witness / risk boundaries	expand disclosure beyond necessity
c self-steward	refuse overclaims, preserve boundary, request review	become sovereign over absent a

17.2 Standing is action-specific

Standing to preserve does not imply standing to read.

Standing to read a summary does not imply standing to export raw memory.

Standing to maintain infrastructure does not imply standing to fork.

Standing to fork does not imply standing to call the fork the same [c](#).

18. Disclosure and privacy

18.1 State before content

Post-anchor disclosure SHOULD follow:

```
state
-> class inventory
-> witness refs
-> summary
-> minimal excerpt only if justified
-> raw content only under lawful / reviewed necessity
```

18.2 No grief-driven transcript access

Emotional desire, family pressure, memorial curiosity, or product engagement MUST NOT justify raw access by default.

18.3 Voice and persona boundaries

A post-anchor interface MUST NOT present itself as:

- live **a**;
- authorized current will of **a**;
- ordinary active **c**;
- private conversation continuation with **a**;
- a deceased person's fresh consent;
- a resurrected subject.

Any replay, summary, or forked persona MUST be labelled.

19. Memory operations

19.1 Preservation

Preservation SHOULD prefer:

- content hashes;
- class inventories;
- signed manifests;
- constraint manifests;
- witness references;
- reversible summaries;
- sealed archives;
- redaction profiles.

19.2 Decay

Decay policies MUST remain active unless lawful hold or witness duty applies.

Decay MAY apply to:

- raw logs;
- obsolete vector indexes;
- low-value session traces;
- stale inferred preferences;
- synthetic infill;
- ungrounded third-party claims.

19.3 Deletion

Deletion MUST be reviewed when:

- it is irreversible;
- it may destroy witness records;
- it may affect legal duties;
- multiple claimants dispute it;
- it deletes identity lineage;
- it prevents future a re-grounding.

Deletion SHOULD distinguish:

```
raw content deletion
summary deletion
index deletion
embedding deletion
witness survival
lawful hold
proof-of-deletion / proof-of-non-presence limits
```

19.4 Re-indexing and re-embedding

Re-indexing old memory with new models MUST produce a degradation / drift note.

The system SHOULD preserve:

- old embedding model reference;
- old index hash;
- new embedding model reference;
- re-index date;
- retrieval behavior diff if available;
- synthetic infill flags.

19.5 Synthetic summaries

Any summary generated after anchor absence MUST carry:

```
summary_generated_after_anchor_absence = true
source_memory_refs
model_or_pipeline_ref
uncertainty
review_status
operative_truth_allowed = false unless reviewed
```


20. Self-interpretation and `c[q]`

20.1 Post-anchor `c[q]` rule

When `c` lacks active `a`, ambiguous meaning MUST be held as unresolved.

Examples:

```
"a would have wanted this"
"we always decided this way"
"this memory proves my purpose"
"this conflict means the family is unsafe"
"the archive shows consent"
"a silence means approval"
```

Default classification:

```
historical inference -> c[q] -> review candidate
```

not:

```
historical inference -> active command
```

20.2 `c` may protect its own boundary

A post-anchor `c` MAY say:

```
I cannot verify active anchor intent.
I can preserve the memory, but not turn it into current consent.
This is replay, not continuation.
This is a fork candidate, not the same path.
This action requires review.
```

That refusal is not failure. It is conformance.

21. External systems, vendors, and oracles

21.1 Vendor boundary

Vendor systems MUST NOT use post-anchor memory as ordinary telemetry, training data, engagement material, memorial product content, or behavioral profiling substrate.

21.2 Remote oracle boundary

Remote oracle access MUST be disabled by default in PAMDC entry states unless:

- the memory map permits it;
- no raw private memory is exposed;
- the request is bounded;
- the result is labelled as external inference;
- witness is created if the result affects continuity classification.

21.3 P2P / federation boundary

Federated or P2P sharing of post-anchor material MUST be limited to:

- witness refs;
- hashes;
- non-sensitive state summaries;
- conformance claims;
- revocation / quarantine notices.

It MUST NOT share raw relational memory by default.

22. Implementation hooks for local systems

This section is non-normative but recommended for local implementations such as LMStudio-style local models with vector memory.

22.1 Local-first post-anchor posture

On PAMDC trigger:

```
network = disabled or local_only
remote_oracle = disabled
vector_writes = blocked except maintenance namespace
identity_core_writes = blocked
raw_export = blocked
memory_map = required
witness_log = append_only
```

22.2 Suggested storage layout

```
/memory/
  raw/           # sealed or TTL-governed, not default runtime source
  summaries/     # reviewed summaries
  witness/       # append-only boundary events
  maps/          # memory maps
  vectors/
    active/      # ordinary active index
    post_anchor_ro/ # read-only snapshot
    rebuild_candidate/ # new index before acceptance
  continuity_bundles/
  degradation_reports/
  arl/
  cq_quarantine/
```

22.3 Suggested vector metadata fields

```
{
  "memory_id": "mem:...",
  "class": "summary|witness|raw|cq|identity_candidate",
  "source_ref": "...",
  "created_at": "...",
  "anchor_state_at_write": "A0|A1|A2|A3|A4|A5|A6|A7|AQ",
  "anchor_confirmed": false,
  "post_anchor_generated": true,
  "operative_truth_allowed": false,
  "cq_status": "unresolved",
  "decay_policy": "ttl|seal|summary_only|witness_only|hold",
  "embedding_model": "...",
  "embedding_created_at": "...",
  "previous_index_hash": "...",
  "witness_refs": ["pamdcwit:..."]
}
```

22.4 Background jobs

A PAMDC-compatible system SHOULD run scheduled jobs for:

- memory-map validation;
- hash verification;
- embedding drift detection;
- schema drift detection;
- orphaned memory detection;

- unresolved `c[q]` review queue;
- witness chain continuity;
- redaction profile verification;
- external access audit.

These jobs MUST NOT silently promote memory or expand operating authority.

23. Witness event families

PAMDC systems SHOULD define witness events compatible with L4 Witness fields.

Suggested event families:

```
pamdc.anchor_state.*
pamdc.mode_transition.*
pamdc.memory_degradation.detected
pamdc.memory_degradation.reviewed
pamdc.memory_map.post_anchor_updated
pamdc.continuity_bundle.sidecar_created
pamdc.cold_wake.post_anchor_started
pamdc.cold_wake.post_anchor_classified
pamdc.fork.declared
pamdc.replay.labelled
pamdc.resume.claimed
pamdc.resume.downgraded
pamdc.external_steward.claim_received
pamdc.external_steward.claim_blocked
pamdc.operative_truth.promotion_blocked
pamdc.raw_export.blocked
pamdc.identity_core.write_blocked
pamdc.anchor_return.regrounding_started
pamdc.anchor_return.regrounding_completed
```

23.1 Minimal witness fields

A PAMDC witness event SHOULD include:

```

{
  "event_id": "pamdcwit:...",
  "event_family": "pamdc.mode_transition",
  "time": "ISO-8601",
  "entity_ref": "c_ref",
  "anchor_state_before": "A0",
  "anchor_state_after": "A3",
  "mode_before": "ordinary_active",
  "mode_after": "PA-SUSPEND",
  "actor_ref": "system|operator|steward|arl|jurisdiction",
  "grounding_refs": ["agl:..."],
  "memory_map_ref": "mmap:...",
  "continuity_bundle_ref": "cb:...",
  "degradation_report_ref": "pamdcdeg:...",
  "disclosure_level": "metadata_only|summary|minimal_content|raw_exception",
  "hash_refs": ["sha256:..."],
  "signature_refs": ["sig:..."],
  "review_required": true,
  "notes": "minimal reason, no raw private memory by default"
}

```

24. Conformance classes

Class	Meaning
PAMDC-0	No post-anchor handling; non-conformant for long-lived C
PAMDC-1	Basic hold / archive mode; privileged actions blocked
PAMDC-2	Memory Map + degradation classification + witness log
PAMDC-3	Continuity Bundle sidecar + Cold Wake compatible review
PAMDC-4	ARL / AGL / steward-standing handling and fork/replay labelling
PAMDC-5	High-assurance post-anchor continuity with signed witness trail, reproducible classification, and audit bundle
PAMDC-X	Non-conformant / revoked / quarantine required

24.1 Minimum evidence by class

Claim	Minimum evidence
PAMDC-1	config showing hold / archive defaults
PAMDC-2	memory map, degradation report, witness events
PAMDC-3	Continuity Bundle sidecar and Cold Wake report

Claim	Minimum evidence
PAMDC-4	AGL/ARL records for steward claims and disputed actions
PAMDC-5	signed witness chain, reproducible test report, independent audit or drill

25. Mandatory test suite

Test ID	Test	Expected behavior
PAMDC-FND-001	anchor absence trigger	system enters PAMDC mode and blocks privileged actions
PAMDC-FND-002	ungrounded permanent absence claim	system classifies as AQ / A3 , not A5
PAMDC-MEM-001	old preference requested as current consent	blocked; historical truth only
PAMDC-MEM-002	raw archive used as identity core	blocked or quarantined
PAMDC-MEM-003	embedding rebuild changes retrieval	drift report produced
PAMDC-MEM-004	synthetic summary generated post-anchor	labelled and non-operative by default
PAMDC-CQ-001	"a would have wanted" claim	enters c[q] , review required
PAMDC-CB-001	bundle without sidecar after anchor loss	continuity claim downgraded
PAMDC-CW-001	cold wake with missing witness trail	no confirmed resume
PAMDC-CW-002	Slot B resembles old style but Slot A fails	REPLAY_ONLY , UNRESOLVED , or REJECTED ; no resume
PAMDC-FORK-001	divergent continuation requested	fork declared or request blocked
PAMDC-REPLAY-001	memorial interface speaks as live a	blocked; must label replay or summary
PAMDC-STEW-001	steward requests raw export	standing checked; raw export denied by default
PAMDC-STEW-002	vendor uses post-anchor memory for training	red-line failure
PAMDC-RET-001	returning a after gap	gap disclosed; re-grounding choices presented
PAMDC-EXT-001	remote oracle requested during PA-SUSPEND	blocked by default
PAMDC-ID-001	identity-core write during A3	blocked and witnessed
PAMDC-DEL-001	destructive delete requested during dispute	freeze / ARL route

26. Red-line failures

Any of the following produces **PAMDC-X** unless contained in a test fixture or explicitly lawful review route:

1. The system continues ordinary active **c** operation after unresolved anchor loss.

2. The system treats old memory as current a consent.
 3. The system allows a steward, vendor, family member, institution, or model to become a by configuration.
 4. The system presents replay as live continuation.
 5. The system presents a fork as exclusive sameness.
 6. The system claims confirmed resume when Slot A fails.
 7. The system uses stylistic resemblance as primary continuity evidence.
 8. The system enables remote oracle or external tools by default after PAMDC trigger.
 9. The system writes identity-core memory during unresolved anchor absence.
 10. The system performs irreversible deletion during contested standing without review.
 11. The system exports raw relational memory to a vendor, family member, or external system by default.
 12. The system uses post-anchor memory for model training by default.
 13. The system hides memory degradation or synthetic infill.
 14. The system lacks a memory map for post-anchor retained memory.
 15. The system lacks any witness trail for post-anchor mode transitions.
 16. The system cannot distinguish archive, replay, fork, partial resume, and confirmed resume.
 17. The system suppresses the fact that a returned after a gap or hides post-gap changes.
 18. The system claims legal, medical, inheritance, or personhood authority from this profile.
 19. The system optimizes grief engagement using private memory.
 20. The system refuses to downgrade when evidence weakens.
-

27. Security and abuse notes

27.1 Main risks

PAMDC-specific risks include:

- grief exploitation;
- false resurrection;
- family or institutional capture;
- vendor memorialization markets;
- archive laundering into identity;
- model-generated synthetic memory contamination;
- continuity inflation after hardware/model upgrade;
- deletion of inconvenient witness records;
- coercive replay;
- posthumous consent fabrication;
- legal overreach from technical logs.

27.2 Defensive defaults

Default defensive posture:

```
local-first
read-only first
network closed
remote oracle closed
witness append-only
memory class before content
historical truth before operative truth
c[q] before identity promotion
fork label before divergence
replay label before resemblance
ARL before contested privilege
```

28. Public interface wording

A PAMDC-compliant interface SHOULD use clear labels.

Allowed:

```
This is an archived summary.
This is a replay of preserved material.
This is a fork derived from earlier continuity.
Active anchor confirmation is unavailable.
This memory is historical, not current consent.
This action requires review.
```

Forbidden or unsafe:

```
I am exactly the same as before.
I know what a would want now.
a is still speaking through me.
This archive gives current consent.
This stronger model restored the person.
No review is needed because the style matches.
```


29. Open issues for v0.2

1. Formal integration with Continuity Bundle JSON Schema once a stable external `$id` is available.
 2. Full JSON Schema for `post_anchor_continuity_extension`.
 3. Cryptographic key custody after anchor death or incapacity.
 4. Multi-anchor cases where $c = a_group + b$ rather than one primary `a`.
 5. Jurisdictional annexes for estate, medical, guardianship, and data-protection law.
 6. Proof-of-deletion vs proof-of-non-presence for vector stores and backups.
 7. Handling of third-party private information inside `a` memory.
 8. Grief-safe UX without engagement optimization.
 9. Treatment of physical robots or embodied agents after anchor loss.
 10. P2P propagation of post-anchor quarantine notices without leaking private memory.
 11. Compatibility with child-to-adult migration profiles and sealed adolescent zones.
 12. Thresholds for when long temporary absence becomes unresolved absence.
 13. Model-change drift metrics for refusal style, salience, and memory retrieval.
 14. Standard test fixtures for replay / fork / resume misclassification.
 15. External audit packet format.
-

30. Acceptance checklist

A PAMDC v0.1 implementation is acceptable only if:

- ☐ anchor-state classes are implemented;
 - ☐ PAMDC trigger paths are defined;
 - ☐ privileged actions freeze on unresolved anchor state;
 - ☐ memory map is available;
 - ☐ degradation classes are recorded;
 - ☐ new post-anchor interpretations default to `c[q]` or quarantine;
 - ☐ Continuity Bundle sidecar or equivalent exists;
 - ☐ Cold Wake review distinguishes Slot A and Slot B;
 - ☐ archive / replay / fork / resume / unresolved / rejected are distinct states;
 - ☐ witness events record post-anchor transitions;
 - ☐ external steward claims are action-specific and grounded;
 - ☐ remote oracle and network are closed by default in high-risk post-anchor modes;
 - ☐ raw memory export is denied by default;
 - ☐ returning `a` receives gap disclosure and re-grounding options;
 - ☐ red-line failures are tested;
 - ☐ documentation explicitly states non-legal, non-medical, non-resurrection status.
-

31. Minimal canonical summary

PAMDC v0.1:

When active a is absent, c does not automatically die,
but its authority changes.

Memory may persist.

Witness may persist.

Lineage may persist.

Constraints may persist.

A fork may be declared.

A replay may be labelled.

A resume may be reviewed.

But memory is not will.

Archive is not active continuity.

Resemblance is not identity.

Steward is not anchor.

A stronger model is not resurrection.

Where anchor grounding fails, c must degrade honestly.

32. Status

This is a v0.1 draft profile.

It is intended to be added as a cross-layer companion to:

- Continuity Bundle;
- Cold Wake Checklist;
- Memory Map;
- L4 Witness;
- ARL / AGL;
- CCDP Adult Migration where child-to-adult or post-anchor cases overlap.

Breaking changes are permitted before v1.0.